

Real-Time Word-Level Temporal Segmentation in Streaming Speech Recognition

Naoto Nishida
nawta@g.ecc.u-tokyo.ac.jp
The University of Tokyo
Tokyo, Japan

Hiroataka Hiraki
hirotakahiraki@gmail.com
The University of Tokyo
Tokyo, Japan
National Institute of Advanced Industrial Science and
Technology
Chiba, Japan

Jun Rekimoto
rekimoto@acm.org
The University of Tokyo
Tokyo, Japan
Sony CSL Kyoto
Kyoto, Japan

Yoshio Ishiguro
ishiy@acm.org
The University of Tokyo
Tokyo, Japan

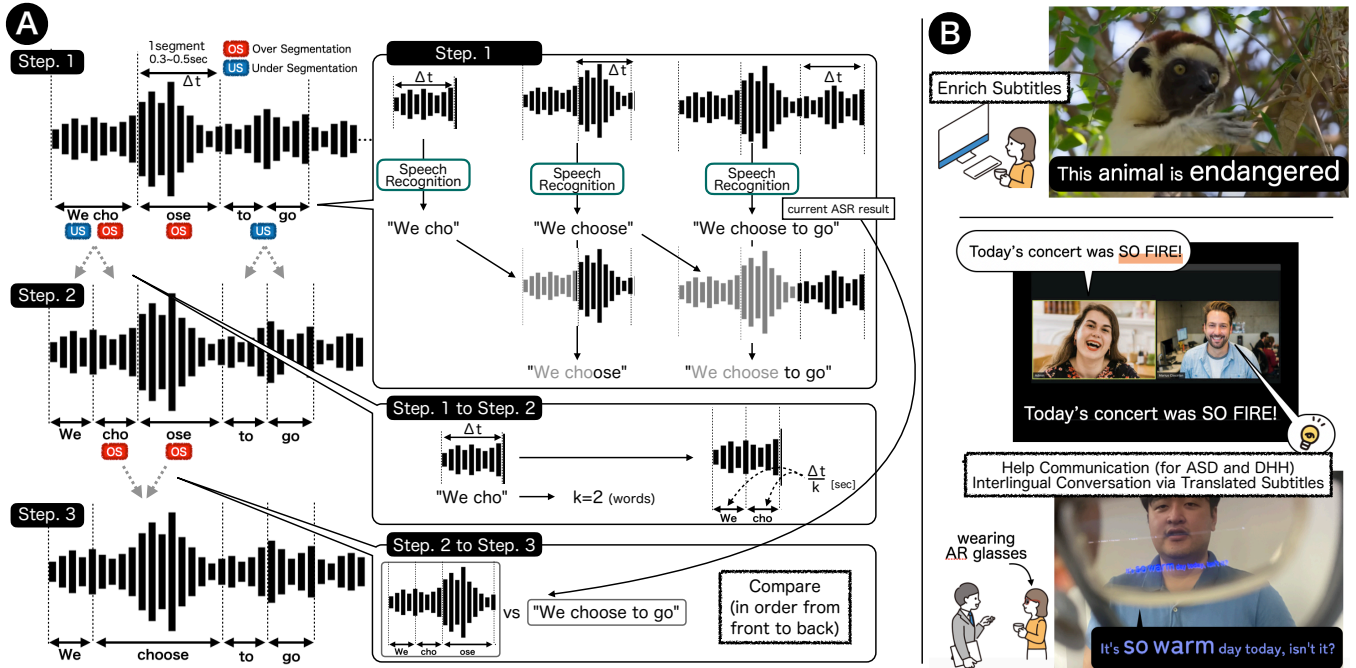


Figure 1: System Workflow and application examples. A) Our system temporally segments streaming Audio Speech Recognition (ASR) results in real time at word level. Workflow in detail is explained in Section 2. B) Our system can be used to add word-level nonverbal information (e.g., volume, tones, and intonations) to texts such as those used for subtitles (upper), and to help communicate with people who find it hard to understand the nuances or people who are communicating via translated texts (middle and lower).

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ABSTRACT

Rich-text captions are essential to help communication for Deaf and hard-of-hearing (DHH) people, second-language learners, and

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those with autism spectrum disorder (ASD). They also preserve nuances when converting speech to text, enhancing the realism of presentation scripts and conversation or speech logs. However, current real-time captioning systems lack the capability to alter text attributes (ex. capitalization, sizes, and fonts) at the word level, hindering the accurate conveyance of speaker intent that is expressed in the tones or intonations of the speech. For example, “YOU should do this” tends to be considered as indicating “You” as the focus of the sentence, whereas “You should do THIS” tends to be “This” as the focus. This paper proposes a solution that changes the text decorations at the word level in real time. As a prototype, we developed an application that adjusts word size based on the loudness of each spoken word. Feedback from users implies that this system helped to convey the speaker’s intent, offering a more engaging and accessible captioning experience.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility; Visualization; Interactive systems and tools.**

KEYWORDS

Speech segmentation, Caption design, Speech accessibility, Paralinguistic cues, Real-time

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1 INTRODUCTION

Rich text captions promise to significantly enhance communication for Deaf and hard-of-hearing (DHH) individuals [3, 6], second language learners [2], and those with autism spectrum disorder (ASD) [9]. By preserving nuances when converting speech to text, these captions can also increase the realism of presentation scripts and conversation logs, making them more engaging and effective. In addition, rich text captions can improve the entertainment value of movies and TV shows, creating a more immersive experience for viewers. The integration of advanced speech-to-text technologies into real-time captioning systems impacts how we communicate and consume media.

However, the problem is that current rich text captioning systems cannot alter text attributes (e.g., capitalizations, sizes, or fonts) at the word level in real-time. This limitation hinders the accurate conveyance of speaker intent, as nuances such as emphasis and intonation are not properly captured. For example, “YOU should do this.” versus “You should do THIS.” illustrates how varying emphasis can change the meaning.

To solve this problem, previous research has focused on developing real-time captioning systems that translate speech into text, but often overlook the importance of capturing and displaying paralinguistic cues. Some systems have attempted to use static text attributes or manual annotations, but these approaches are either not dynamic or require extensive manual input, making them impractical for real-time applications [4, 6]. Other tools such as whisper

series [5, 10, 11] need to spend relatively longer time to recognize (e.g., a few seconds to 30 seconds) the speech if it is used alone. There remains a gap in developing a system that can dynamically alter word-level text attributes based on real-time speech analysis. To fill in this gap, we design and implement a real-time rich-text captioning system that performs word-level temporal segmentation and timestamping using streaming speech recognition. As shown in Fig. 1, our system takes an input speech stream, segments it into words, assigns timestamps to each word, and dynamically adjusts the text decorations based on the detected paralinguistic cues such as loudness and emphasis. Specifically, the system works in three steps: first, it segments the audio signal according to frame rates, which can vary; second, it segments these frames further based on spaces or morphological analysis; third, it compares these segments with the latest Audio Speech Recognition (ASR) results to extract accurate timestamps and recognized words.

The main contribution of this paper is the development of a real-time rich-text captioning system that enhances the conveyance of speaker intent through dynamic word-level text decorations. In contrast to prior work that relied on static or manually annotated captions, our approach leverages advanced speech recognition technologies to provide a more engaging and accessible communication experience for a diverse range of users.

2 SYSTEM OVERVIEW

As in Step 1, our system segments the audio signal to recognize the speech (illustrated as dotted lines in Figure 1). The interval of each segment varies depending on the computers’ capacity and conditions. It then recognizes the speech in the segment, where some may be over-segmented (ex. ‘ose’ in at step 1 in Fig 1), under-segmented (ex. ‘to go’), or both occur (ex. ‘We cho’). To enhance the recognition accuracy, the system uses concatenated audio segments from the beginning to the newest segment. From step 1 to step 2, it segments under-segmented segments according to space (or the results from morphological analysis in languages without spacing). We linearly divided Δt into k segments, each of which consists of $\Delta t/k$ seconds. Therefore, all segments have over-segmented segments or properly segmented segments in step 2. From step 2 to step 3, it compares step 2’s result with the newest ASR result, which has proper grammatical segmentation information (which is spacing) to concatenate over-segmented parts (ex. ‘cho’ and ‘ose’ at step 2 in Fig. 1). As a result, our system extracts the segments of audio signals, their corresponding recognized words, and their timestamps of when the speech started and ended, illustrated at step 3 in Fig. 1. As the practical implementation detail, we used Unity to display the decorated subtitles. Therefore, the program was coded in C# on the Unity platform. We also used Azure Speech Recognition API [1] for ASR tasks.

3 DISCUSSION

Although promising, several challenges that have emerged during the use require further improvement. First, the current integration of the over-segmented portions (Step 2 to Step 3) operates with a time complexity of $O(n^2)$, which is inefficient. Future work should focus on optimizing this step to enhance computational efficiency.

Second, our method for handling the sub-segmentation by dividing segments based on the word count per time step is just one approach. There is an inevitable error. Exploring alternative algorithms could improve accuracy and performance towards this error. Third, additional applications for the system should be explored, such as real-time translation, educational tools for second language learners, and communication aids for ASD individuals. User studies will help validate these applications. Valuable past works would also help us design how to integrate nonspeech information in audio signals into captions to build useful applications for these people in our future work [8]. Fourth, addressing the delay in real-time processing is crucial. Minimizing lag will ensure natural communication flow. Fifth, we currently use fixed timestamps for speech segments. Predicting the average duration of the speech from initial words and backcalculating start times could improve timing accuracy. Sixth, our current integration method depends on the lag between steps. If processing speeds increase, this method may become unstable. Developing a more robust integration method is necessary. Lastly, for languages without word spacing (e.g., Japanese), additional segmentation (e.g., McCab [7]) is needed after Step 1. Further user studies will validate the application examples and provide feedback for further refinement. Addressing these challenges will enhance the efficiency, accuracy, and versatility of our system.

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